KVIS HPC SHOWCASE · HPC & BRAIN CAVE

# AI HPC
# AI Research for Thai Stocks with HPC

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· PROJECT IN BRIEF

## AI
## The AI practises decisions and learns from feedback

AI 5 - PPO, A2C, DDPG, TD3 SAC -

The project compares five AI learning methods on Thai-stock data using one controlled process that measures return, risk and trading costs.

| 503 | 5 | 29.7% |
|---|---|---|
| / planned runs | / AI methods | |

HPC · WHERE HPC HELPS

## HPC
## HPC makes large experiments faster and checkable

HPC

A laptop is used for coding and small tests; HPC schedules repeated runs, records job IDs and resources, and preserves evidence for reproduction.

**CODE → SCHEDULE → RUN → VERIFY**

/ SOFTWARE · Python · Stable-Baselines3 · Gymnasium · Optuna · SLURM

· FROM CODE TO EVIDENCE

### 1 · / CODE

5

```
for method in METHODS:
    train(method)
    verify(manifest, receipt)
```

### 2 · HPC / SCRIPT

```
#SBATCH --cpus-per-task=2
#SBATCH --mem=8GB
#SBATCH --time=02:00:00
#SBATCH --partition=compute-devel
```

3

ADD TEAM PHOTO

3

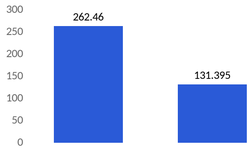
3 STUDENT RESEARCHERS

### 3 · / RUN

JOB 3550 · / COMPLETED

00:33:20 · AI 5
2 CPUs · 809,680 KB peak RAM

### 4 · / RESULT




-49.94%

5 manifests + 5 receipts
0 checkpoints

HPC · TEAM PERSPECTIVE

### / NOTH

HPC

HPC lets us repeat the same idea while keeping a job record we can check.

### / MICKEY

Good computing preserves essential evidence without wasting storage.

: Job 3550 · : 150
**Verified: job 3550 completed · Next: recheck capacity before the 150-trial screening**

17 . . 2026 · full screening ·